COVID-19 Diagnosis Prediction Under Noisy Labels

Fixing the Data Before Fixing the Model

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Team **Unemployed_Statisticians** | Final Rank: **#2** | Private Score: **0.99130**

The Task and the Twist

Binary classification: predict whether a patient tested positive for COVID-19 from symptoms, demographics, and clinical indicators.

Dataset at a glance

- ∼1,500 training rows, ∼3,500 test rows
- 18 features: 3 continuous, 2 categorical, 13 binary
- Balanced classes (778 pos. / 722 neg.)
- `comorbidity` 53% NaN, recoded as "None"

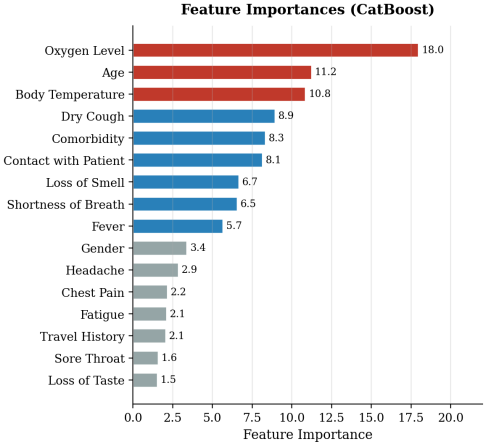
The catch

- We were told a subset of training labels had been *deliberately flipped*
- Roughly **19% of labels are wrong**
- Models trained on noisy labels plateaued at 74% to 80%
- Scored on **accuracy**: you get the label right or you do not

Label noise shaped our entire approach.

Data Overview and Feature Importance

- `oxygen_level` dominates (importance ≈ 18)
- `age` and `body_temperature` are next
- Key symptoms: `dry_cough`, `shortness_of_breath`, `loss_of_smell`
- Many binary features contribute under 5% each but collectively matter



CatBoost feature importances (exploratory).

Why Standard Models Fail Here

The core issue: with ∼19% label noise, any model that fits training labels hard will learn the noise.

Model	CV Accuracy	Problem
CatBoost (Optuna-tuned)	0.7487	Memorises noisy labels
LightGBM	0.7673	Same issue
LDA	0.7567	Linear, underpowered
QDA	0.7167	High variance, small n
GaussianNB	0.7013	Assumes feature independence
k-NN (k=31)	0.7047	Noise in neighbours
Logistic Regression (T1+T2)	0.9147	Decent after label correction
TabPFN 5-seed (T1+T2)	**0.9460**	**Best by far**

- All tree-based and classical models hit 0.70 to 0.77 on *noisy* labels. No tuning broke through.
- Even LR jumps to 0.91 once labels are fixed. The bottleneck was **label quality**, not model complexity.

What is TabPFN?

TabPFN (Hollmann et al., 2022) is a transformer *meta-trained* (trained to learn from new tasks) on millions of synthetic tabular classification datasets.

How it works

- The entire training set is fed as *context* in a single forward pass, similar to prompting an LLM
- Predictions via **in-context learning**: no gradient updates, no fine-tuning on your data
- Approximates the Bayesian posterior predictive: $P(y_{\text{test}} \mid X_{\text{test}}, X_{\text{train}}, y_{\text{train}})$

Why it fits our task

1. Pre-training = regularisation
Cannot overfit to 1,500 rows since it never trains on them via gradient descent

2. Noise robustness
Bayesian posterior smooths over individual flipped labels

3. No hyperparameters
Zero risk of overfitting to CV folds

Noise Detection: Three Independent Methods

Instead of relying on any single noise detection technique, we ran three methods that use *completely different* signals and combined their outputs.

Method 1: Confident Learning (Cleanlab). *Probabilistic.*
5-fold OOF logistic regression $\rightarrow$ Cleanlab estimates joint distribution of noisy vs. true labels.
Flagged **285** samples (19.0%).

Method 2: Area Under the Margin (AUM, CatBoost). *Margin-based.*
Compute margin $m_i(t) = P_t(\text{true class}) - P_t(\text{other class})$ at 5 CatBoost checkpoints (100 to 500 iters).
Samples with mean AUM < 0.40 are consistently hard-to-learn. Flagged **352** samples.

Method 3: Dataset Cartography. *Training-dynamics.*
5 seeds $\times$ 5 folds of LR $\rightarrow$ mean per-sample confidence $P(\text{true label})$.
Hard-to-learn threshold: mean confidence < 0.45. Flagged **302** samples.

One method is probabilistic, one is margin-based, one is training-dynamics-based.
What matters is where they agree.

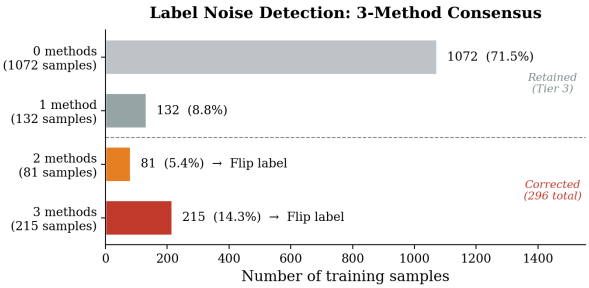
Consensus Tiers and Label Correction

Voting rule: count how many of 3 methods flagged each sample.

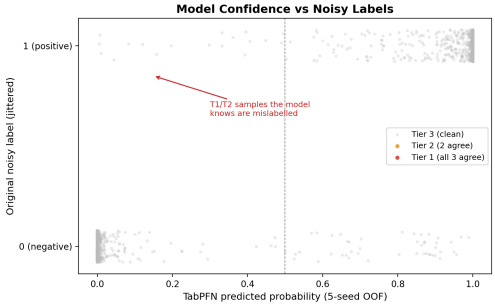
- **Tier 1** (all 3 agree): **215** samples
  → Label flipped. Near-certain noise.
- **Tier 2** (2 of 3 agree): **81** samples
  → Label flipped. Probable noise.
- **Tier 3** (0 or 1 method): **1,204** samples
  → Label retained as given.

Total corrected: 296 labels.

This is a precision/recall tradeoff on noise detection itself. We chose precision: flipping a correct label is just as damaging as keeping a wrong one.



Impact: Before vs. After Label Correction



TabPFN OOF probabilities vs. original labels. Clusters in the lower-right and upper-left are the corrected samples.

Progression of private scores:

Strategy	Private
S1: T1 only	0.99015
S2: +Features	0.98261 ↓
S3: T1+T2	0.99073
S4: 5-seed	**0.99130**

- S1→S3: Tier 2 corrections give **+0.00058**
- S2: feature engineering *hurt* (−0.00754)
- S3→S4: 5-seed ensemble adds **+0.00057**
- 10 total submissions, no leaderboard probing

The 5-Seed Ensemble: Variance Reduction

Problem: TabPFN has stochastic internal processing. Different random seeds produce slightly different probability estimates.

Solution: Train with 5 seeds (42, 123, 314, 777, 2026), average the probability vectors.

Pipeline per seed:

1. 5-fold stratified CV $\rightarrow$ OOF probabilities
2. Full-train model $\rightarrow$ test probabilities
3. Repeat for all 5 seeds

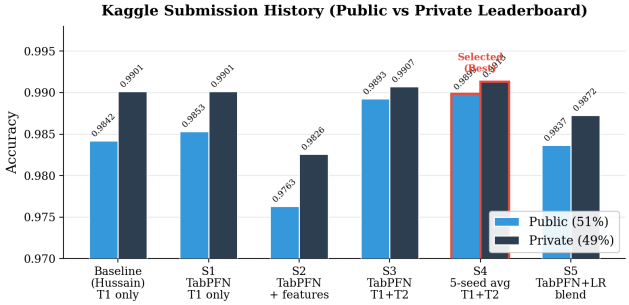
$\Rightarrow$ 25 OOF estimates per training sample
$\Rightarrow$ 5 test prediction vectors, averaged

Why this works:

- This is *not* diversity-based ensembling (blending different model types)
- It is pure **variance reduction** over the same architecture
- Blending TabPFN with LR actually *hurt* (0.9220 vs. 0.9460). Diluting a strong model with a weaker one reduces signal.

Final prediction: classify at threshold 0.50. Threshold sweeping on OOF showed no meaningful gain.

Competition Trajectory



- Only **10 total submissions**; deliberate, not leaderboard-probing
- Each submission tested exactly one hypothesis
- Private (0.99130) $>$ Public (0.98985)
  $\rightarrow$ **no overfitting to public split**
- S2 (feature eng.) was the only regression, confirming TabPFN's prior $>$ hand-crafted features
- Final rank: **#2 overall**

Key Takeaways

1. **Label quality matters more than model complexity.**
   Every model trained on noisy labels hit a ceiling of 74% to 80%. After cleaning 296 labels, we hit 99%+.
2. **Consensus-based noise detection is safer than any single method.**
   Three independent signals agreeing on the same sample is strong evidence. Voting filters false positives.
3. **Pre-trained transformers can work on tabular data, under the right conditions.**
   Small n, binary classification, clean(ed) labels. In-context learning avoids overfitting.
4. **Feature engineering can hurt when the model's prior is already stronger.**
   On 1,500 rows, TabPFN's inductive bias outperformed manual interactions.
5. **Ensemble the same strong model; do not blend with weaker ones.**
   Multi-seed averaging reduces variance. Blending with LR diluted the signal.

Conclusion

Private Score: 0.99130 | Rank: #2

What we did

- 3-method noise detection pipeline
- 296 label corrections (215 T1 + 81 T2)
- TabPFN v2, 5-seed ensemble
- No feature engineering, no threshold tuning
- 10 principled submissions

What mattered most

- The jump from $\sim$0.75 CV accuracy to 0.99 on Kaggle came from **fixing labels**
- Clean data did the heavy lifting, not model choice
- Private $>$ Public confirms no overfitting

Future work

- Pseudo-labelling of confident test predictions
- Extending correction into Tier 3

Code: `https://github.com/n33levo/unemployed_statistician`